

Dynamic and Adversarially Robust Graph Learning for Adaptive Cybersecurity in Evolving Cyber–Physical Systems

Mohammad Ryiad Al-Eiadeh

Modern cyber–physical systems (CPS), including power grids, industrial control systems, transportation infrastructures, distributed energy resources, and enterprise networks, operate in environments where vulnerabilities, dependencies, and adversarial behaviors continuously evolve. Components may join or leave the network, vulnerabilities may emerge or disappear, and attackers may adapt their strategies over time. These evolving conditions create highly dynamic attack surfaces that challenge traditional cybersecurity planning and defense optimization approaches. Attack graphs provide a natural representation for modeling attack propagation, privilege escalation, and dependency relations across interconnected systems. Consequently, graph-based cybersecurity has become an important direction for intelligent security resource allocation and automated cyber defense. However, most existing graph-theoretic and learning-based approaches assume static graph structures during optimization and analysis. In practice, this assumption becomes increasingly unrealistic for modern large-scale infrastructures where network conditions evolve continuously.

My proposals aims to develop a unified graph-learning framework for adaptive and adversarially robust cybersecurity in evolving CPS. In particular, the proposed work focuses on three tightly connected research directions: dynamic two-level node embeddings for evolving attack graphs, cluster-based multi-defender security resource allocation, and adversarial attacks on graph representations and embedding spaces. The broader objective is to bridge graph representation learning, scalable optimization, adversarial machine learning, and automated cybersecurity decision-making to support intelligent and resilient defense systems for large-scale critical infrastructures.

A central component of this work is the development of a dynamic two-level graph representation framework capable of capturing both local structural evolution and global risk propagation behavior in evolving attack graphs. The first level focuses on local structural embeddings that model neighborhood-level changes caused by node additions, node removals, edge modifications, and evolving attack paths. This component investigates scalable graph representation learning techniques such as random walks, node2vec, struct2vec, Graph Neural Networks (GNNs), and transformer-based graph encoders. These methods preserve short-range structural dependencies and local attack-surface dynamics while adapting efficiently to graph updates.

The second level focuses on global diffusion embeddings that capture long-range dependency and attack propagation patterns across interconnected infrastructures. While local structural changes are important, many cybersecurity risks propagate indirectly through distant graph pathways and interdependent system components. To model these behaviors, the proposed framework investigates diffusion-based methods such as Personalized PageRank and Log Personalized PageRank (Log-PPR), which capture how vulnerabilities and attacks influence remote nodes throughout the network. The local and global embedding spaces will then be integrated using low-rank matrix approximation and embedding alignment techniques, including Singular Value Decomposition (SVD). Instead of recomputing graph embeddings from scratch after every network update, the framework incrementally updates only affected graph regions, significantly improving scalability and computational efficiency.

Another major direction of this research is extending traditional single-defender cybersecurity models into a cluster-based multi-defender framework. Most existing attack-graph defense strategies assume a centralized defender responsible for protecting the entire infrastructure. However, real-world systems are often managed by multiple organizations or operational entities that independently secure different regions of the network. To address this limitation, the proposed framework partitions attack graphs into localized security regions using embedding-driven clustering methods such as K-Means, spectral clustering, and graph community detection. Each defender secures a localized subgraph while coordinating globally to minimize overall system risk and social security cost. Resource allocation decisions are guided by graph embeddings, attack propagation behavior, node importance, and inter-cluster dependencies. This direction aims to support scalable and distributed cybersecurity planning for large interconnected infrastructures.

A further objective of this research is to investigate adversarial attacks targeting attack-graph topology and graph embedding representations. As graph-learning-based cybersecurity systems become increasingly automated, attackers may intentionally manipulate graph structures to mislead defense mechanisms and distort cybersecurity decision-making. Such adversarial manipulations include node removal, edge perturbation, phantom asset injection, asset-label flipping, and topology distortion. These attacks may alter node rankings, embedding spaces, clustering structures, and automated defense allocation strategies. Beyond structural manipulation, attackers may also poison embedding representations or diffusion behaviors to influence latent graph representations directly. The proposed research investigates how these adversarial perturbations degrade defense performance and explores robust graph-learning mechanisms capable of maintaining stable and trustworthy cybersecurity decisions under adversarial conditions.

The resulting graph representations will be integrated into an automated cybersecurity decision-making framework capable of predicting near-optimal defense allocation strategies for evolving attack graphs. Instead of solving expensive optimization problems after every graph update, the proposed system learns to infer effective defense strategies directly from dynamic graph representations. This approach enables real-time adaptive cybersecurity planning while significantly improving scalability and operational efficiency. Potential predictive models include Factorization Machines, XGBoost, neural networks, and graph-based learning architectures.

The proposed framework will be evaluated using both synthetic and realistic attack-graph datasets representing cyber-physical systems and interdependent infrastructures. Experimental studies will compare the proposed methods against static graph embedding approaches, full graph recomputation methods, and existing graph-theoretic defense strategies. Performance will be assessed in terms of computational scalability, embedding update efficiency, adaptability to evolving graph structures, robustness against adversarial perturbations, quality of defense allocation strategies, and reduction in overall system risk and security cost.

Overall, this research aims to contribute a scalable two-level dynamic embedding framework for evolving attack graphs, a cluster-based multi-defender cybersecurity architecture, and adversarially robust graph-learning mechanisms for automated cyber defense. More broadly, the proposed work seeks to strengthen the theoretical and computational foundations of trustworthy AI-driven cybersecurity for large-scale interconnected systems operating under uncertainty and adversarial conditions. Beyond cybersecurity, the same modeling principles may be extended to infrastructure resilience analysis, cascading-failure systems, biological networks, and other large-scale dynamic graph domains.